

PREDICTING STUDENT EMPLOYABILITY USING MACHINE LEARNING BASED ON SOFT SKILLS ASSESSMENT

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Abstract

Employability is vital for graduates to succeed in competitive job markets and reflects higher education institutions' effectiveness. It is essential to investigate which specific traits contribute to a higher success rate of employability, as understanding these factors can help optimize targeted interventions and improve employment outcomes. This research aims to identify and analyze the key traits that influence student employability using educational data mining techniques integrated with machine learning. It also seeks to provide an explainable framework that can help improve job readiness among graduates. The study utilizes a dataset collected from a survey of 2,982 students, where soft skills are rated on a scale of one (1) to five (5). By applying predictive modeling, the research explores the role of machine learning in career forecasting, including predicting job suitability and identifying skill gaps.

Keywords: Soft Skills, Employability, Machine Learning, SEMMA, Decision Tree, Support Vector Machine, K Nearest Neighbor

1. Introduction

Employability has become a critical concern for graduates as they face increasingly competitive and uncertain job markets, and it also reflects how effective higher education institutions are in preparing students for real-world work. Identifying the specific traits that contribute to employability is important, as this can guide targeted interventions and improve employment outcomes. Employment rate is the extent to which available labour resources are being used. Employed people are those aged 15 or over who report that they have worked in gainful employment for at least one hour in the previous week. The working age population refers to people aged 15 to 64. Recent global reports, such as the 2026 Employment and Social Trends report of the International Labour Organization, show that although unemployment rates remain stable, many workers still lack access to decent and secure jobs, with millions experiencing poverty and informal employment.

In the Philippines, the Department of Labor and Employment (DOLE) reported that unemployment rose to 5.8 percent in January 2026, from 2.17 million to 2.96 million, affecting nearly three million Filipinos and highlighting the ongoing challenges in the labor market. These conditions emphasize the need for graduates to possess not only technical knowledge but also strong soft skills that can improve their chances of employment.

Machine learning is a branch of artificial intelligence that uses statistical and computational techniques to develop systems that improve through data-driven experience. In the context of mental health, machine learning has shown promise in predicting risk levels and identifying vulnerable groups using available demographic and behavioral data (Jordan & Mitchell, 2015). Commonly used algorithms such as decision trees, logistic regression, and random forests can be applied to classify individuals into risk categories and provide actionable insights (Dabek & Caban, 2015).

This study aims to predict student employability levels—employable and less employable—based on their soft skills. By applying a structured machine learning approach, (TBD), the study seeks to identify patterns and build predictive models. The dataset consists of 2,982 students, where soft skills are rated on a scale of one to five. The research will evaluate model performance using metrics such as accuracy, confusion matrix, and class-specific measures to determine how well soft skills can predict employability and identify potential skill gaps.

2. Related Literature

In recent years, many researchers attempted to use machine learning in higher education to enhance graduate's features and curricula to support employability (Saidani et al., 2021). Machine learning has been increasingly applied in mental health research to predict risk and identify individuals who may need intervention. Garriga et al. (2022) developed a predictive model using electronic health record (EHR) data to forecast mental health crises. Their model achieved an AUROC of 0.797 for predicting a crisis within 28 days and demonstrated practical value by helping prevent crises in 19% of cases. This study highlights how structured clinical data can be effectively leveraged with machine learning algorithms to support timely interventions and improve mental health outcomes, emphasizing the potential of predictive analytics in healthcare settings.

ElSharkawy et al. (2022) conducted a study to predict IT graduates' employability using a dataset of 296 graduates. They applied the Decision Tree (DT) algorithm using an 80% training and 20% testing split, with features such as trainings, soft skills, hard skills, and industry-demanded skills. The Decision Tree model, which uses a tree-like structure based on IF-THEN rules and the Gini index for splitting, achieved the highest performance among all models. It obtained 100% accuracy, precision, recall, and F1 score, and produced the highest number of correct predictions with minimal errors. The results indicate that Decision Tree is highly effective in identifying patterns between students' skills and their employability outcomes.

In the same study on predicting IT graduates' employability, SVM was applied alongside other machine learning models using features such as trainings, soft skills, and industry-demanded skills. The dataset was divided into 80% training and 20% testing, and the model's performance was evaluated using accuracy, precision, recall, and F1 score. The results showed that SVM achieved a high accuracy of 98%, making it one of the top-performing models next to the Decision Tree. It also demonstrated strong precision and F1 score, indicating its effectiveness in correctly classifying employable and non-employable graduates. Although it did not outperform the Decision Tree model, SVM still provided reliable and consistent results. This suggests that SVM is a suitable method for predicting student employability, especially when dealing with complex relationships between soft skills and labor market requirements.

In another study, presented in the IOP Conference Series by Saini et al. (2021): Materials Science and Engineering, K-Nearest Neighbors (KNN) was applied as one of the classification techniques to predict students' employability and expected salary range based on selected academic and skill-related attributes. The study emphasizes that the effectiveness of KNN depends heavily on the distance metric used in determining proximity among data points. Using

a dataset of 262 student records, the results showed that KNN achieved an accuracy of 81.01% in predicting employability and 77.21% for salary range classification, which is relatively lower compared to other algorithms like Random Forest. Despite this, KNN still demonstrated its capability in handling classification problems in educational data mining, particularly due to its simplicity and ability to work well with structured datasets

3. Research Question

This study seeks to examine the role of soft skills in predicting student employability using machine learning techniques. It focuses on assessing how various soft skills contribute to the classification of students as Employable or Less Employable based on collected survey data. Specifically, it seeks to answer the following questions:

1. Is there a significant relationship between soft skills and student employability?
2. Which soft skills significantly influence the prediction of student employability?
3. Which data mining algorithm provides the highest accuracy in predicting student employability?

4. Materials and Method

4.1 Data Source and Method

The data used in this study was obtained from Kaggle open source platform. It contains mock job interview results collected from various university agencies in the Philippines. The dataset consists of 2,982 observations, where each record represents an individual student assessed based on soft skills during the interview process. The data is structured using a 5 point Likert scale, where 1 indicates the lowest and 5 indicates the highest level of each assessed skill. It includes eight independent variables related to soft skills and one dependent variable that classifies student employability as Employable or Less Employable.

This study follows a SEMMA framework to guide the data mining. SEMMA is a data mining methodology developed by SAS that stands for Sample, Explore, Modify, Model, and Assess. Below is the illustration of SEMMA.

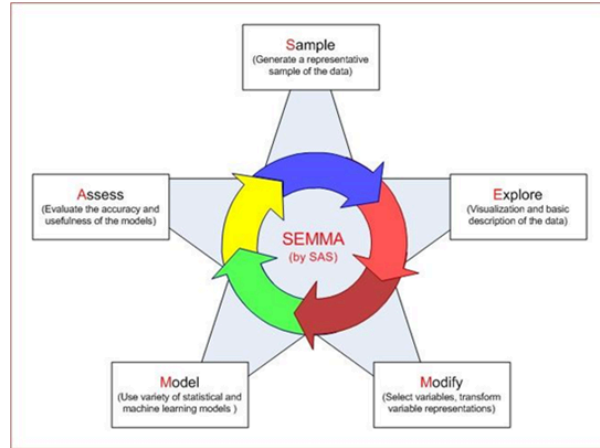


Figure 1. Illustration of SEMMA Process Model

SEMMA intends to make it easy to apply exploratory statistical and visualization techniques, select and transform the most significant predictive variables, model the variables to predict outcomes, and finally confirm a model's accuracy (Olson & Delen, 2008).

The SEMMA method guided the workflow.

Sample

This phase focuses on selecting the dataset to be used in the study. The dataset consists of 2,982 student responses collected through a survey on soft skills. The data is divided into training and testing sets using a 70 percent and 30 percent split. The training set is used to build the models, while the testing set is used to evaluate their performance.

Explore

In this phase, the dataset is examined to understand its structure and characteristics. Descriptive statistics and frequency distributions are analyzed for each variable. The study also checks the balance between the Employable and Less Employable classes. This step helps identify patterns, trends, and potential issues such as data imbalance or anomalies that may affect model performance.

Modify

This phase prepares the data for modeling. The Likert scale responses are already numerical, but further preprocessing is applied. The target variable is encoded into binary form. Feature scaling is performed for algorithms like Support Vector Machine and k Nearest Neighbors to ensure consistent value ranges. Any missing or inconsistent data is handled during this stage. This step improves the quality of the input data and enhances model accuracy.

Model

In this phase, machine learning algorithms are applied to build predictive models. The study uses

three classification algorithms — Decision Tree, Support Vector Machine, and k Nearest Neighbors. Each model is trained using the training dataset. These algorithms are selected because they are widely used for classification tasks and allow comparison of different learning approaches.

Assess

This final phase evaluates the performance of the models. A confusion matrix is used to measure how well each model classifies students into Employable or Less Employable. Performance metrics such as accuracy, precision, recall, and F1 score are computed. The results of all models are compared, and the model with the best performance is selected as the final predictive model.

4.2 Exploratory Data Analysis

Exploratory Data Analysis is conducted to examine the distribution and relationships of the variables in the dataset before applying machine learning models. The dataset consists of 2,982 student responses with eight independent variables and one target variable, all measured using a 5 point Likert scale. This step ensures that the data is suitable for analysis and provides an initial understanding of patterns that may influence employability prediction.

To analyze the distribution of responses, pie charts are generated for each variable. These charts present the proportion of responses for each Likert scale value from 1 to 5, expressed in percentages. This visualization allows a clear view of how students rated themselves across different soft skills and how responses are distributed within each feature.

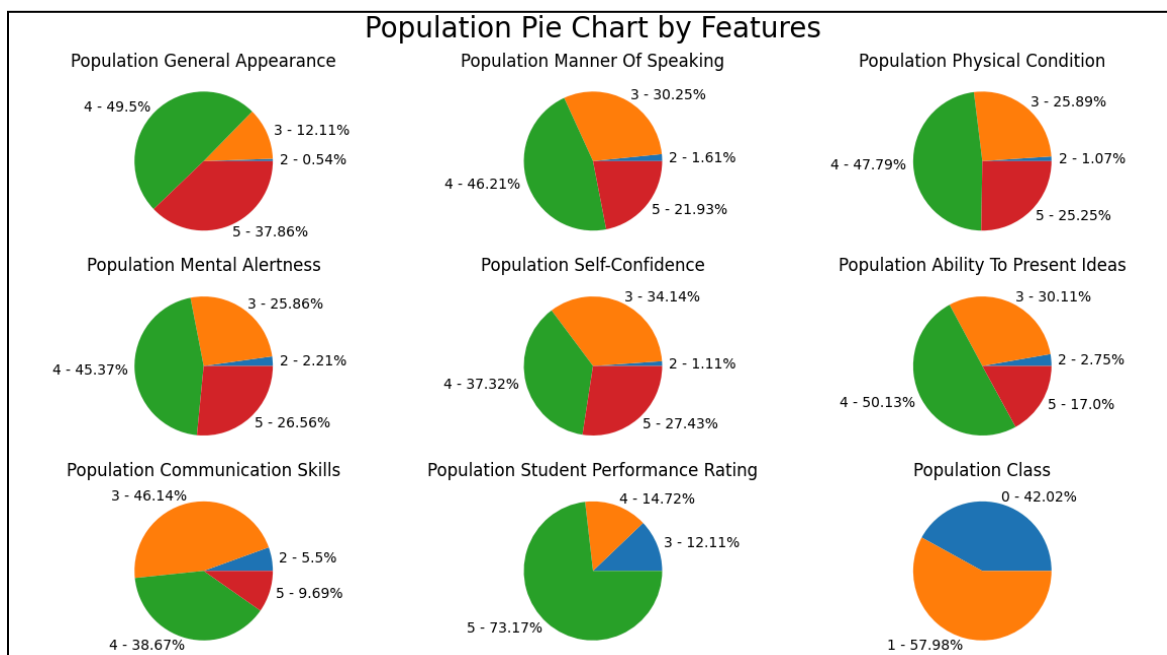


Figure 2. Distribution of Responses Across Variables

The pie chart analysis shows that most variables have responses concentrated in the higher range of the Likert scale, particularly in values 3, 4, and 5. This suggests that students generally rate themselves at a moderate to high level across the measured soft skills. Some variables show a more balanced distribution across all scale values, indicating variability among student responses. This variation is important for machine learning, as it allows models to detect patterns and differences between students. However, variables with responses heavily concentrated in a single rating may provide limited discrimination power and are noted during modeling.

The distribution of the target variable also shows the proportion of Employable and Less Employable classes. This helps determine whether the dataset is balanced. If one class dominates, it may affect model performance and require careful interpretation of evaluation metrics such as precision, recall, and F1 score.

To further examine the relationships among variables, a correlation analysis is performed using Spearman correlation. Since the dataset is based on a Likert scale, which represents ordinal data, Spearman correlation is more appropriate as it evaluates relationships based on ranked values rather than assuming equal intervals between scale points. This method is effective in identifying monotonic relationships between variables.

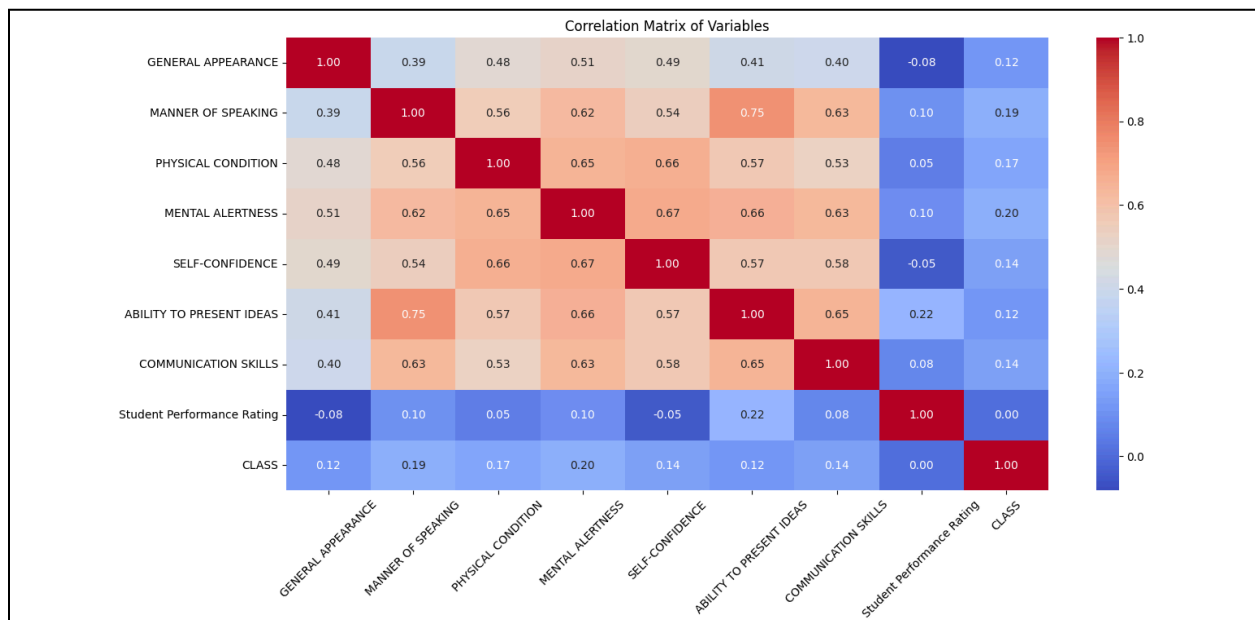


Figure 3. Spearman Correlation Matrix of Variables

The Spearman correlation matrix shows the relationships among the independent variables and their association with the target variable. Most soft skill variables exhibit moderate positive correlations with each other, with values generally ranging from 0.40 to 0.67. This indicates that these skills tend to increase together. For example, Mental Alertness shows a strong relationship with Self Confidence (0.67) and Physical Condition (0.65), while Communication Skills is moderately correlated with Ability to Present Ideas (0.65) and Manner of Speaking (0.63). The strongest observed relationship is between Manner of Speaking and Ability to Present Ideas (0.75), which suggests that students who communicate well verbally are also effective in

presenting ideas. Despite these moderate to strong relationships, the correlation values do not exceed 0.80, which indicates that multicollinearity is not a serious concern. Each variable still captures a distinct aspect of student soft skills, so all features are retained for modeling.

In contrast, Student Performance Rating shows weak correlations with most soft skills, with values close to zero and some slightly negative values such as with General Appearance (-0.08) and Self Confidence (-0.05). This suggests that performance rating does not strongly align with the other soft skill measures and may represent a different dimension of student evaluation.

When examining the relationship with the target variable, all independent variables show weak positive correlations with the Class variable, with values ranging from 0.12 to 0.20. Mental Alertness (0.20) and Manner of Speaking (0.19) show the highest associations with employability, although the relationships remain weak. This indicates that no single variable strongly determines employability on its own. Instead, employability is likely influenced by the combined effect of multiple soft skills.

Overall, the correlation analysis suggests that while soft skills are moderately related to each other, their individual relationships with employability are weak. This supports the use of machine learning models, as these models can capture complex patterns and interactions among multiple variables rather than relying on simple linear relationships.

4.3 Data Preprocessing

Data preprocessing is performed to ensure that the dataset is clean and appropriate for machine learning analysis. To avoid bias, duplicate records are removed and missing values are checked and changed. All responses are validated to ensure that it falls between 1 and 5 on the Likert scale. The target variable is encoded into numerical form, where Employable is assigned 1 and Less Employable is assigned 0. Feature scaling using Min Max normalization is applied to standardize the range of independent variables, which is important for algorithms such as Support Vector Machine and k Nearest Neighbors. The dataset is then split into training and testing sets using a 70 to 30 ratio to evaluate model performance on unseen data.

4.4 Machine Learning Models

For this study, these following predictive modeling techniques are utilized:

1. Decision Tree

Decision Tree is a supervised machine learning algorithm that is used for both classification and regression tasks. It represents a decision using a tree structure, where internal nodes refer to the attributes, the branches represent the decision rules, and leaf nodes represent the final class labels or the outcomes. This structure makes it simple to interpret and visualize the data by allowing the model to split it into smaller subsets based on the feature values (Aurélien Geron, 2022). This algorithm is widely used in predictive modeling since it is easy to understand and interpret. It makes it possible for researchers to identify the factors that most affect the result. Decision trees are appropriate for analyzing a variety of inputs because they can handle both numerical and categorical data (Susmita Ray, 2019) making them suitable for identifying students

employability. In educational data analysis, the Decision Trees are effective because they provide clear decision rules that can be easily explained. Because of this, it can be used to support data-driven decisions in education and to predict student employability. They are suitable for this study because of their interpretability and flexibility. It is given with the following equation:

Gini impurity:

$$\text{Gini} = 1 - \sum(p_i^2)$$

where p_i is the proportion of samples belonging to class i inside a node.

The model chooses splits that reduce this value.

2. Support Vector Machine

Support Vector Machine is a supervised learning algorithm developed by Vladimir Vapnik and Corinna Cortes in 1995 to handle classification, regression, and pattern recognition problems. It is widely used for analyzing both linear and non-linear relationships (Subaira et al., 2014; Trstenjak et al., 2014). It classifies objects into categories (above or lower plane) based on the characteristics of the object and using the Kernel techniques it can transform nonlinear to linear before partitioning (Adejo & Connolly, 2018). When it comes to multiple predictive variables, SVM performs well, particularly when group prediction is preferred over individual prediction. SVM has been applied in different domains such as finance, healthcare, and marketing. In the field of education, several studies have demonstrated its effectiveness in predicting student employability. For instance, in the study of Casuat et al. (2020) found that the Support Vector Machine (SVM) algorithm was the most accurate in predicting graduate employability with a 91.22% accuracy rate. These results show that the SVM is a reliable and accurate model for tasks such as educational prediction. Usually, the SVM regression function can be formulated as follows:

$$y = f(x) = w \cdot \varphi(x) + b$$

Where:

$\varphi(x)$: High-dimensional feature space that is nonlinearly mapped from the input space x

y : Estimated value in terms of input data x

3. K Nearest Neighbor

K-Nearest Neighbors is a simple classification algorithm first proposed by Evelyn Fix and Joseph Hodges in 1951 and further developed by Thomas Cover in 1967 (Balderas et al., 2021). This algorithm is a supervised machine learning algorithm that is mostly used for classification purposes. The algorithm consists of a variable parameter, known as k , which translates to the number of 'nearest neighbours'. The KNN algorithm functions by finding the nearest data

point(s) or neighbour(s) from a training dataset for a query. The nearest data points are found according to the closest distances from the query point. After locating the k nearest data points, it performs a majority voting rule to find which class appeared the most. The class that appeared the most is ruled to be the final classification for the query (Gide et al., 2022). Several studies have demonstrated the effectiveness of kNN in classifying student work readiness. For example, a study by Alfian and Lhaksmana (2024) applied kNN to a classifying student work readiness and achieved high prediction accuracy, reaching up to 97.50% percent. This indicates that kNN is capable of determining whether a student is employable or less employable. In this study, kNN is utilized to classify students' employability based on their soft skills. One of the most popular choices to measure this distance is the Euclidean distance (Hansun & Tanuwijaya, 2019) given with the equation:

$$D(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

Where:

D : Distance between p and q

n : Training set

i : index i th

Once the value of k is selected, predictions can now be performed. For regression, the predictions are the average of the k nearest neighbors' outcome. It is given with the formula:

$$y = \frac{1}{k} \sum_{i=1}^k y_i$$

Where:

y : Predicted value

k : Number of Neighbors

y_i : Value at index i

5. Results

5.1 Model Performance

This section presents the performance of three machine learning models used to predict student employability based on soft skills. The models include Decision Tree, Support Vector Machine, and k Nearest Neighbors. Each model was evaluated using a 70 percent training set and 30 percent testing set. Performance was evaluated using confusion matrix analysis and standard classification metrics.

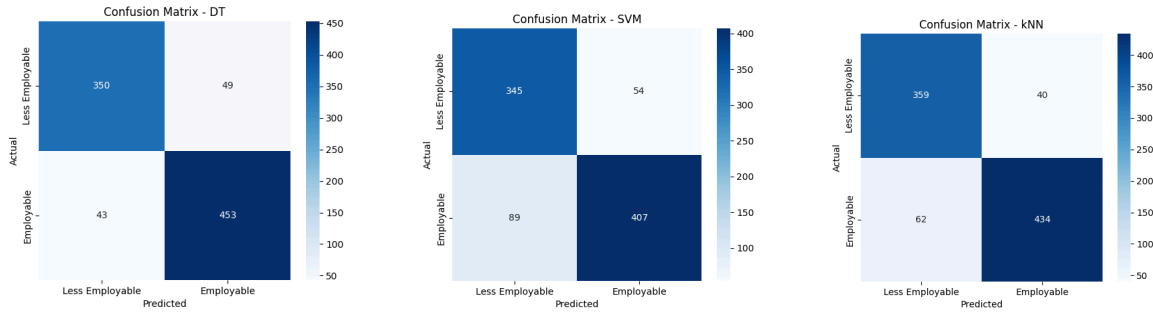


Figure 4. Confusion Matrix Graph of Each Model

Figure 4 presents the confusion matrices for each classification model, while Table 1 provides a detailed summary of their classification outcomes. The confusion matrix illustrates how each model classifies students into Employable and Less Employable categories by showing correct and incorrect predictions.

In a standard confusion matrix layout, the top left cell represents true negatives, the top right represents false positives, the bottom left represents false negatives, and the bottom right represents true positives.

Table 1. Confusion Matrix Table

MODEL	TRUE NEGATIVE	FALSE POSITIVE	FALSE NEGATIVE	TRUE POSITIVE
DT	350	49	43	453
SVM	345	54	89	407
KNN	359	40	62	434

The Decision Tree model demonstrated strong classification performance. It correctly identified 453 Employable students and 350 Less Employable students. However, it misclassified 49 Less Employable students as Employable and 43 Employable students as Less Employable. This indicates that the model maintains a good balance between both classes with relatively low misclassification. The Support Vector Machine model using the RBF kernel showed moderate performance. It correctly classified 407 Employable students and 345 Less Employable students. However, it produced a higher number of false negatives at 89, meaning many Employable students were incorrectly classified as Less Employable. This suggests that while the model is capable of separating classes, it is more conservative in predicting Employable outcomes. The k Nearest Neighbors model achieved a balanced performance across both classes. It correctly identified 434 Employable students and 359 Less Employable students. It also recorded the lowest false positives at 40 among all models, indicating that it is more reliable when predicting Employable students. Although it has slightly higher false negatives than the Decision Tree, it maintains overall stability in classification. Overall, the Decision Tree and kNN models show

strong classification capability, while the SVM model demonstrates comparatively weaker performance due to higher misclassification of Employable students.

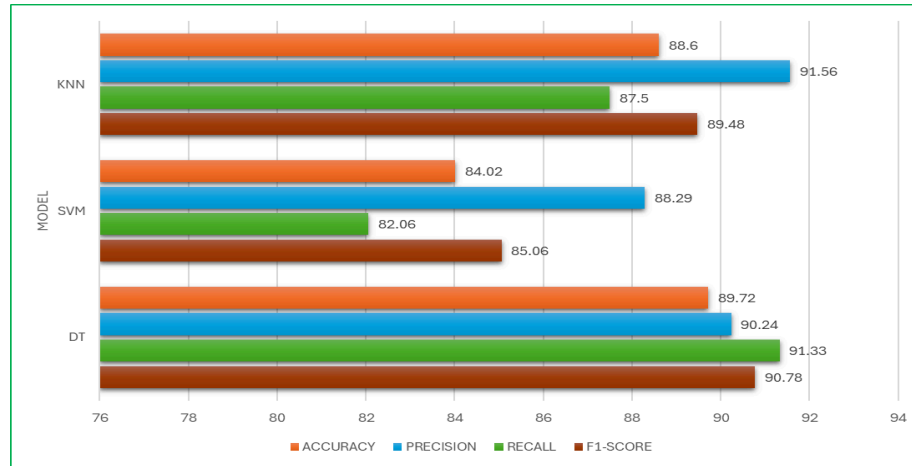


Figure 5. Assessment Scores for Each Model

The Decision Tree model achieved the highest accuracy at **89.72** percent, indicating strong overall predictive performance. It also shows a high recall of **91.33** percent, which means it is effective in identifying Employable students. Its balanced precision and recall values result in the highest F1 score among the three models. The k Nearest Neighbors model follows closely with an accuracy of **88.60** percent. It achieved the highest precision at **91.56** percent, indicating that when the model predicts a student as Employable, it is highly reliable. Its recall is slightly lower than the Decision Tree, but it still maintains a strong balance between correctly identifying both classes. The Support Vector Machine model achieved the lowest performance among the three models, with an accuracy of **84.02** percent. Although its precision remains relatively high, its recall is lower compared to the other models, indicating that it misses more Employable students during classification. Based on these results, the Decision Tree model demonstrates the best overall performance due to its highest accuracy and balanced evaluation metrics. The kNN model also shows competitive performance and may be considered as an alternative depending on the desired balance between precision and recall.

5.2 Model Interpretation

This section presents the interpretation of the selected model using SHAP analysis to explain how each soft skill influences the prediction of student employability. A SHAP beeswarm plot was generated to identify the contribution and impact of each independent variable on the model's output. The features are arranged based on their importance, where those at the top have the greatest influence on the prediction.

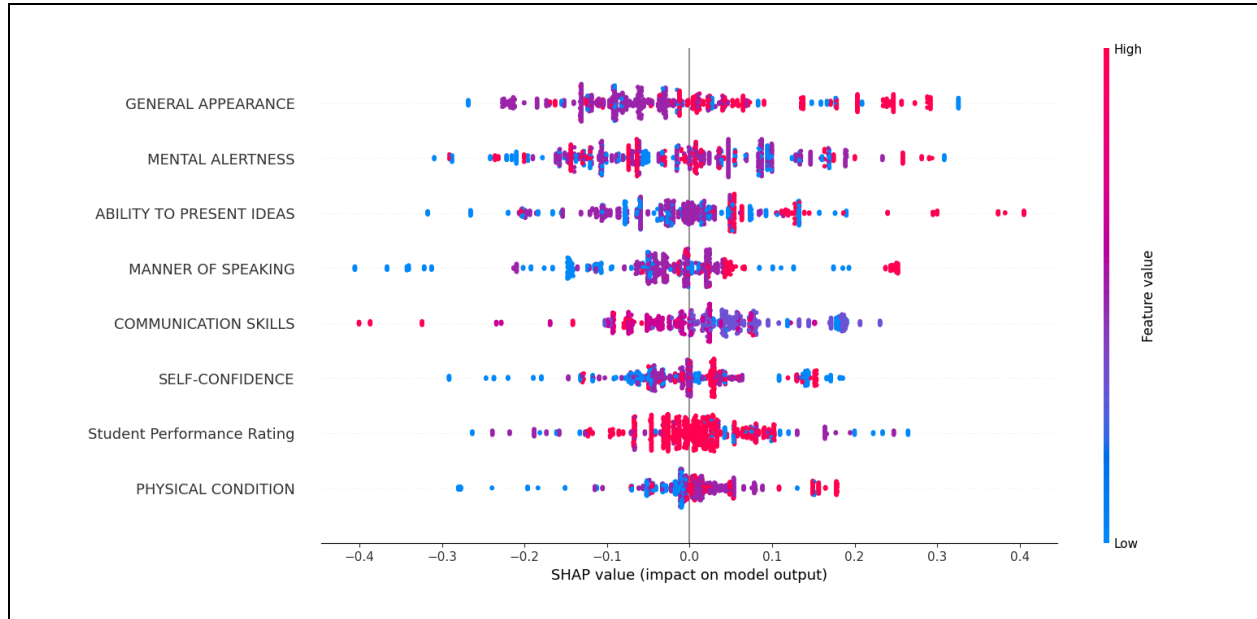


Figure 5. SHAP Model Interpretation

The results show that General Appearance is the most influential feature in the model. Higher values of this variable are associated with positive SHAP values, indicating a strong contribution toward predicting students as Employable. Mental Alertness follows as another significant factor, where students with higher alertness levels are more likely to be classified as Employable, while lower levels push predictions toward Less Employable. The Ability to Present Ideas also plays a major role, as higher scores consistently increase the likelihood of employability classification.

Manner of Speaking and Communication Skills are also identified as key contributors. Students who demonstrate better speaking manner and strong communication skills tend to have predictions shifted toward Employable. These findings indicate that communication related competencies are critical in determining employability outcomes. Self Confidence shows a moderate level of influence, where higher confidence slightly improves the probability of being classified as Employable.

On the other hand, Student Performance Rating and Physical Condition exhibit lower impact compared to other variables. Although they still contribute to the model, their influence is less significant in determining the final classification. This suggests that while academic performance and physical attributes are relevant, they are not the primary factors driving employability predictions in this study.

Overall, the SHAP analysis reveals that employability is largely influenced by communication ability, mental alertness, and presentation skills. Higher levels of these soft skills increase the likelihood of being classified as Employable, while lower levels shift predictions toward Less Employable. This supports the idea that employers prioritize cognitive and communication related competencies when assessing student employability.

6. Discussion and Findings

6.1 Interpretation and Findings

The findings of this study show that soft skills play a significant role in predicting student employability. Based on the model results, the Decision Tree algorithm achieved the highest accuracy at **89.72** percent, followed by k Nearest Neighbors at **88.60** percent, and Support Vector Machine at **84.02** percent. This indicates that tree-based models are more effective in capturing patterns within the dataset, especially when dealing with structured and interpretable features such as soft skills.

The confusion matrix results further support this observation. The Decision Tree model produced a balanced classification with relatively low false positives and false negatives. This means the model can reliably distinguish between Employable and Less Employable students. In comparison, the Support Vector Machine model showed a higher number of false negatives, indicating that it tends to misclassify Employable students as Less Employable. The k Nearest Neighbors model demonstrated stable performance and achieved the highest precision, which suggests that its positive predictions are highly reliable.

The SHAP analysis provides deeper insight into which variables influence employability prediction. The results reveal that General Appearance, Mental Alertness, and Ability to Present Ideas are the most significant predictors. These variables consistently contribute to positive classification outcomes when their values are high. This implies that students who demonstrate strong presentation skills, quick thinking, and professional appearance are more likely to be classified as Employable.

Communication-related variables such as Manner of Speaking and Communication Skills also show strong influence. These findings highlight that employers value how students express ideas and interact with others. Self Confidence contributes moderately, indicating that confidence supports employability but is not the primary deciding factor.

On the other hand, Student Performance Rating and Physical Condition have lower contributions in the model. This suggests that academic-related measures and physical attributes are less influential compared to communication and cognitive skills. The result supports the idea that employability is more dependent on soft skills than purely academic performance.

Overall, the findings confirm that there is a meaningful relationship between soft skills and employability. The results also answer the research questions by identifying key predictors and determining that the Decision Tree model provides the best performance among the selected algorithms.

6.2 Conclusion

This study aimed to predict student employability using machine learning based on soft skills assessment. The results demonstrate that machine learning models can effectively classify students into Employable and Less Employable categories using soft skill data.

Among the models used, the Decision Tree algorithm achieved the best overall performance due to its high accuracy and balanced evaluation metrics. The k Nearest Neighbors model also showed strong performance, while the Support Vector Machine model performed slightly lower due to higher misclassification rates.

The study also identified the most important soft skills influencing employability. General Appearance, Mental Alertness, and Ability to Present Ideas were found to be the strongest predictors. Communication Skills and Manner of Speaking also played a major role. These findings indicate that communication and cognitive abilities are critical factors in determining employability outcomes.

The study concludes that soft skills are strong indicators of employability and can be effectively analyzed using machine learning techniques. The results can help educators and institutions design targeted interventions to improve student readiness for employment.

6.3 Limitations

This study has several limitations that should be considered. First, the dataset was obtained from a secondary source, which limits control over how the data was collected. The responses are based on self-reported or assessment-based ratings, which may introduce bias.

Second, the dataset is limited to soft skills measured using a Likert scale. Other important factors such as technical skills, work experience, and academic specialization were not included. This may affect the completeness of employability prediction.

Lastly, the study only used three machine learning algorithms. Although these models provide strong results, other advanced models such as Random Forest or other machine learning models that could potentially improve prediction performance.

6.4 Future Work

Future studies can improve this research by incorporating additional variables such as technical skills, internships, academic performance, and industry certifications. Including these factors may provide a more comprehensive prediction of employability.

Researchers can also explore advanced machine learning techniques such as ensemble models to improve accuracy and robustness. Comparing more algorithms can help identify the most effective approach for this type of problem. Another area for future work is the use of real-time or longitudinal data. Tracking student development over time can provide deeper insights into how employability evolves and which interventions are most effective.

Finally, future research can focus on developing a practical system or application based on the model. This system can be used by schools to assess students and provide recommendations for improving their employability skills.

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Appendix

#EDA

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_excel(r"C:\Users\msu\Downloads\CleanData.xlsx")

import math

# Gets the piechart vaariables
def pie_chart_var(col_name, df):
    counts = [g.shape[0] for _, g in df.groupby(col_name)]
    headers = [g.loc[:, col_name].iloc[0] for _, g in df.groupby(col_name)]
    # get percentages
    percentages = np.array(counts) / np.array(counts).sum() * 100
    headers = [f"{label} - {round(percentage, 2)}%" for label, percentage in zip(headers, percentages)]
    return counts, headers

# Creating the figure
col = 3
row = math.ceil(len(df.columns) / col)
fig, ax = plt.subplots(row, col, figsize=(20, 10), layout='constrained')
for idx, ax in enumerate(ax.flatten()):
    if idx < len(df.columns):
        col_name = list(df.columns)[idx]
```

```

        counts, headers = pie_chart_var(col_name, df)

        ax.pie(counts, labels=headers)

        ax.set_title(f"Population {col_name.title()}")

    else:

        ax.remove()

fig.suptitle("Population Pie Chart by Features", fontsize=20)

plt.show()

```

#Correlation Matrix

```

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

corr_matrix = df.corr(method='spearman')

plt.figure(figsize=(10, 8))

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")

plt.title("Correlation Matrix of Variables")

plt.xticks(rotation=45)

plt.yticks(rotation=0)

plt.tight_layout()

plt.show()

```

#DT

```

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.read_excel(r"C:\Users\msu\Downloads\CleanData.xlsx")

print(df.head())

from sklearn.model_selection import train_test_split

X = df.drop('CLASS', axis=1)

y = df['CLASS']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(random_state=42)

```

```

dt_model.fit(X_train, y_train)

y_pred = dt_model.predict(X_test)

from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score,
f1_score

cm = confusion_matrix(y_test, y_pred)

accuracy = accuracy_score(y_test, y_pred)

precision = precision_score(y_test, y_pred)

recall = recall_score(y_test, y_pred)

f1 = f1_score(y_test, y_pred)

print("Confusion Matrix:\n", cm)

print("Accuracy:", accuracy)

print("Precision:", precision)

print("Recall:", recall)

print("F1 Score:", f1)

importance = pd.Series(dt_model.feature_importances_, index=X.columns)

print(importance.sort_values(ascending=False))

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.metrics import confusion_matrix

labels = ['Less Employable', 'Employable']

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 5))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',

            xticklabels=labels,

            yticklabels=labels)

plt.title('Confusion Matrix - DT')

plt.xlabel('Predicted')

plt.ylabel('Actual')

plt.show()

```

#SVM

```

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

```

```

from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score

df = pd.read_excel(r"C:\Users\msu\Downloads\CleanData.xlsx")

# print(df.head())

X = df.drop('CLASS', axis=1)

y = df['CLASS']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

svm_model = SVC(kernel='rbf', class_weight='balanced', C=1, gamma='scale', random_state=42)

svm_model.fit(X_train, y_train)

y_pred = svm_model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

accuracy = accuracy_score(y_test, y_pred)

precision = precision_score(y_test, y_pred)

recall = recall_score(y_test, y_pred)

f1 = f1_score(y_test, y_pred)

print("Confusion Matrix:\n", cm)

print("Accuracy:", accuracy)

print("Precision:", precision)

print("Recall:", recall)

print("F1 Score:", f1)

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.metrics import confusion_matrix

labels = ['Less Employable', 'Employable']

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 5))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',

            xticklabels=labels,

            yticklabels=labels)

plt.title('Confusion Matrix - SVM')

plt.xlabel('Predicted')

plt.ylabel('Actual')

```

```
plt.show()
```

#KNN

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score

df = pd.read_excel(r"C:\Users\msu\Downloads\CleanData.xlsx")
X = df.drop('CLASS', axis=1)
y = df['CLASS']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train, y_train)
y_pred = knn_model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Confusion Matrix:\n", cm)
print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.metrics import confusion_matrix

labels = ['Less Employable', 'Employable']
```

```

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 5))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',

            xticklabels=labels,

            yticklabels=labels)

plt.title('Confusion Matrix - kNN')

plt.xlabel('Predicted')

plt.ylabel('Actual')

plt.show()

```

#SHAP

```

import pandas as pd

import shap

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

df = pd.read_excel(r"C:\Users\msu\Downloads\CleanData.xlsx")

X = df.drop('CLASS', axis=1)

y = df['CLASS']

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=42

)

dt_model = DecisionTreeClassifier(random_state=42)

dt_model.fit(X_train, y_train)

explainer = shap.Explainer(dt_model, X_train)

shap_values = explainer(X_test)

# Select class (Employable)

shap_values_class1 = shap_values[:, :, 1]

shap.plots.beeswarm(shap_values_class1)

```